

Product Requirements Document (PRD)

Project Title

AutoAssist: Intelligent OBD-II Vehicle Telemetry Anomaly Detection and Health Monitoring Platform

1. Executive Summary

AutoAssist is an automotive analytics platform designed to analyze OBD-II vehicle telemetry data and identify abnormal operating behavior using a combination of automotive diagnostic principles, telemetry analytics, and machine learning.

The system processes vehicle sensor streams, learns normal operating patterns, detects anomalous behavior, and presents vehicle health insights through an interactive dashboard.

The platform focuses on offline analysis of recorded OBD-II telemetry datasets while maintaining a scalable architecture capable of supporting future real-time telemetry ingestion from OBD-II adapters and connected vehicle systems.

The objective is not to replace OEM diagnostics or ECU fault detection systems, but to provide an intelligent telemetry analysis layer capable of identifying unusual operating patterns, emerging degradation trends, and abnormal sensor relationships.

2. Problem Statement

Traditional OBD-II readers primarily provide:

- Raw sensor values
- Diagnostic Trouble Codes (DTCs)
- Basic live telemetry

These tools rely on ECU-defined thresholds and typically identify issues only after predefined fault conditions are met.

There is limited capability for:

- Detecting subtle deviations in sensor behavior
- Identifying emerging anomalies before DTC generation
- Understanding multivariate relationships between vehicle sensors
- Accounting for varying driving contexts
- Providing health-oriented telemetry analytics

AutoAssist addresses this gap by applying machine learning and automotive domain logic to telemetry streams, enabling deeper interpretation of vehicle operating behavior.

3. Vision

To create an intelligent vehicle telemetry analytics platform capable of learning normal vehicle behavior, identifying anomalies, generating interpretable health insights, and supporting future predictive maintenance workflows using OBD-II sensor data.

4. Product Goals

Primary Goals

- Analyze OBD-II telemetry datasets
 - Learn normal vehicle operating patterns
 - Detect anomalous sensor behavior
 - Generate vehicle health insights
 - Visualize telemetry trends and anomalies
 - Provide interpretable anomaly scores
 - Minimize false-positive anomaly detection through driving-context awareness
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Secondary Goals

- Support multiple driving conditions
 - Enable long-term vehicle behavior analysis
 - Create a scalable backend architecture
 - Prepare for future live telemetry integration
 - Establish reusable telemetry analysis pipelines
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5. Target Users

Automotive Enthusiasts

Users interested in understanding vehicle behavior beyond traditional fault codes.

Students and Researchers

Individuals studying:

- Automotive analytics
 - Vehicle diagnostics
 - Time-series analysis
 - Machine learning applications in transportation
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Fleet and Maintenance Concepts

Future adaptation for:

- Fleet health monitoring
 - Predictive maintenance research
 - Vehicle performance analytics
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6. Scope

In Scope

Telemetry Processing

- OBD-II dataset ingestion
- Dataset validation
- Dynamic schema mapping
- Data cleaning
- Missing value handling
- Sensor normalization
- Timestamp processing

Analytics

- Sensor trend analysis
- Correlation analysis
- Driving condition analysis
- Operational state analysis
- Statistical feature extraction

Machine Learning

- Baseline learning
- Anomaly detection
- Behavioral pattern analysis
- Vehicle health scoring
- Explainable anomaly interpretation

Visualization

- Interactive dashboards
 - Sensor plots
 - Anomaly visualizations
 - Health indicators
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Out of Scope

- ECU tuning

- Vehicle control systems
 - ECU firmware modification
 - Safety-critical decision making
 - OEM diagnostic replacement
 - Guaranteed fault diagnosis
 - Autonomous vehicle control
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7. Functional Requirements

FR-1 Dataset Upload

The system shall allow users to upload OBD-II telemetry datasets in CSV format.

Input

CSV telemetry files.

Output

Validated telemetry records available for processing.

Validation Requirements

The system shall verify:

- Required telemetry fields are available
- Sufficient data exists for baseline generation
- Timestamp continuity is acceptable
- Dataset quality exceeds minimum thresholds

If baseline requirements are not met, the system shall notify the user and prevent unreliable anomaly assessment.

FR-2 Data Preprocessing

The system shall:

- Detect missing values
- Remove invalid records
- Normalize numerical features
- Validate timestamps
- Standardize telemetry schemas
- Prepare telemetry data for analysis

Schema Standardization

The system shall map telemetry fields from various OBD-II logging formats into a canonical internal schema.

Example standardized fields:

- rpm
- speed
- throttle_position
- engine_load
- coolant_temperature
- intake_air_temperature
- maf
- fuel_trim
- timestamp

This mapping layer shall enable compatibility across multiple telemetry sources without modifying downstream analytics pipelines.

FR-3 Telemetry Analysis

The system shall:

- Analyze sensor behavior over time
- Generate descriptive statistics
- Identify significant trends
- Detect sensor relationships
- Evaluate operating characteristics

Examples:

- RPM behavior
 - Vehicle speed patterns
 - Engine load variations
 - Temperature trends
 - Throttle response characteristics
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FR-4 Operational State Classification

The system shall classify telemetry into operational driving states before anomaly analysis.

Example states include:

- Idle
- Low-speed urban driving
- Highway cruising
- Acceleration

- Deceleration
- Stop-and-go traffic

The classification output shall be used to provide contextualized anomaly detection and reduce false-positive alerts.

FR-5 Feature Engineering

The system shall derive analytical features including:

- Rolling averages
- Sensor variability metrics
- Sensor relationships
- Trend indicators
- Rate-of-change calculations
- Context-aware operational features
- State-specific behavioral indicators

Examples:

- RPM fluctuation rates
 - Load-to-speed relationships
 - Throttle response characteristics
 - Acceleration consistency
 - Temperature stability metrics
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FR-6 Anomaly Detection

The system shall identify unusual operating behavior based on learned telemetry patterns.

Anomaly evaluation shall be performed relative to the detected operational state rather than a single global baseline.

Outputs:

- Normal
- Warning
- Abnormal

Each result shall include an anomaly score.

FR-7 Baseline Confidence Assessment

The system shall maintain a confidence score representing the quality and completeness of the learned baseline.

The confidence score shall consider:

- Dataset size
- Operational state coverage
- Sensor completeness
- Data quality

Example outputs:

- Baseline Established
- Limited Baseline Confidence
- Insufficient Data

The anomaly detection engine shall use this confidence level when generating assessments.

FR-8 Explainable Anomaly Interpretation

The system shall provide interpretable explanations for detected anomalies.

For each anomaly event, the platform shall identify major contributing telemetry features.

Example:

- RPM variability
- Engine load deviation
- Throttle inconsistency
- Temperature abnormalities

The architecture shall support future integration of advanced explainability techniques for deeper model interpretation.

FR-9 Vehicle Health Assessment

The system shall generate an overall vehicle health indicator derived from telemetry analysis.

Example:

- Healthy
- Monitor
- Attention Required

Health assessment shall consider:

- Anomaly frequency
 - Anomaly severity
 - Sensor stability
 - Behavioral consistency
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FR-10 Visualization Dashboard

The system shall display:

Vehicle Metrics

- RPM
- Speed
- Engine load
- Throttle position
- Temperature-related sensors
- Airflow-related sensors

Analytics

- Time-series charts
 - Correlation visualizations
 - Operational state visualizations
 - Anomaly markers
 - Health summaries
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FR-11 Reporting

The system shall provide:

- Anomaly summaries
 - Telemetry insights
 - Operational state summaries
 - Health assessment results
 - Contributing factor analysis
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8. Machine Learning Requirements

Initial Approach

Unsupervised anomaly detection.

Recommended models:

- Isolation Forest
- One-Class SVM
- DBSCAN

Model selection may evolve based on dataset characteristics.

Expected Outputs

Anomaly Score

Continuous score representing deviation from learned behavior.

Status Classification

- Normal
 - Warning
 - Abnormal
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Baseline Confidence

Confidence level associated with learned behavior patterns.

Contributing Factors

Identification of telemetry features contributing to abnormal behavior.

9. System Architecture

Frontend

Technology:

- React Application

Responsibilities:

- Dashboard rendering
 - File upload
 - Data visualization
 - User interaction
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Backend

Technology:

- FastAPI

Responsibilities:

- API layer
 - File handling
 - Data validation
 - Schema mapping
 - Data processing orchestration
 - Model execution
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Analytics Engine

Technology:

- Python

Libraries:

- Pandas
- NumPy
- Scikit-learn

Responsibilities:

- Data preparation
 - Operational state classification
 - Feature engineering
 - Baseline learning
 - Anomaly detection
 - Health scoring
 - Explainability generation
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Visualization

Technology:

- Plotly

Responsibilities:

- Interactive telemetry charts
 - Analytical visualizations
 - Operational state displays
 - Anomaly overlays
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10. Non-Functional Requirements

Performance

- Process datasets efficiently
 - Support large telemetry files
 - Maintain responsive dashboard performance
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Scalability

Architecture shall support future additions including:

- Real-time telemetry ingestion
 - Continuous monitoring
 - Multi-vehicle analysis
 - Cloud deployment
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Maintainability

- Modular backend design
 - Separated analytics layer
 - Reusable machine learning pipeline
 - Extensible schema mapping framework
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Reliability

- Graceful handling of incomplete datasets
 - Robust anomaly detection workflow
 - Consistent API responses
 - Reliable baseline confidence assessment
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11. Future Expansion Roadmap

The platform architecture shall remain compatible with future enhancements.

Potential future capabilities include:

Live OBD-II Integration

- ELM327 support
- OBDLink support
- Bluetooth telemetry acquisition

- Wi-Fi telemetry acquisition
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Real-Time Monitoring

- Continuous telemetry streaming
 - Live anomaly detection
 - Live health monitoring
 - Streaming analytics
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Vehicle Profiling

Support for:

- Vehicle-specific baselines
 - Engine-specific operating profiles
 - Vehicle-class normalization
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Historical Analysis

- Long-term vehicle tracking
 - Trend analysis
 - Behavioral drift detection
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Predictive Maintenance

Potential future research areas:

- Battery degradation indicators
 - Cooling system stress indicators
 - Airflow system anomalies
 - Fuel efficiency degradation trends
 - Sensor degradation analysis
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Fleet Analytics

Future support for:

- Multiple vehicles
 - Centralized monitoring
 - Comparative health scoring
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12. Success Metrics

The project will be considered successful if it can:

- Process OBD-II telemetry datasets successfully
 - Establish reliable operational baselines
 - Detect meaningful anomalous behavior
 - Reduce context-related false positives
 - Generate interpretable anomaly scores
 - Explain contributing anomaly factors
 - Visualize telemetry effectively
 - Deliver actionable vehicle health insights
 - Support future migration toward live OBD-II telemetry monitoring
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13. Technology Stack

Frontend:

- React

Backend:

- FastAPI

Analytics:

- Python

Data Processing:

- Pandas
- NumPy

Machine Learning:

- Scikit-learn

Visualization:

- Plotly

Storage:

- CSV Datasets

Model Persistence:

- Joblib